ASSIGNMENT – 2

JAVA PROGRAMMING

User-Defined Exception Handling

Name: Geona Alis J

Register No.: 192511088

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CO Reference: CO2

Assignment Set: Direct-Descriptive

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1. PROBLEM UNDERSTANDING

The objective of this program is to demonstrate the use of a user-defined exception in Java.

A custom exception named InvalidAgeException is created by extending the built-in Exception class. The program validates a person's age and considers an age below 0 or above 120 as invalid.

When an invalid age is entered, the custom exception is thrown and caught using a try-catch block. An appropriate error message is then displayed to the user.

2. APPROACH / DESIGN

The program follows these steps:

① Create a custom exception class called InvalidAgeException.

② Extend the Java Exception class to create the user-defined exception.

③ Create a validateAge() method to check whether the entered age is valid.

④ If the age is less than 0 or greater than 120, throw InvalidAgeException.

⑤ Use a try-catch block in the main() method to handle the exception.

⑥ Display a suitable message for both valid and invalid inputs.

✧ Validation Rule ✧

Valid Age → $0 \leq Age \leq 120$

Invalid Age → $Age < 0$ OR $Age > 120$

3. SOURCE CODE

```
import java.util.Scanner;

class InvalidAgeException extends Exception {
    public InvalidAgeException(String message) {
        super(message);
    }
}

public class Main {

    static void validateAge(int age) throws InvalidAgeException {
        if (age < 0 || age > 120) {
            throw new InvalidAgeException(
                "Invalid age: Age must be between 0 and 120."
            );
        }
```

```
        System.out.println("Valid age: " + age);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter age: ");
        int age = sc.nextInt();

        try {
            validateAge(age);
        } catch (InvalidAgeException e) {
            System.out.println(e.getMessage());
        }

        sc.close();
    }
}
```

✦ Concepts Demonstrated ✦

extends Exception → Creates a custom exception
throw → Throws the exception
throws → Declares the exception
try-catch → Handles the exception
Scanner → Accepts user input

4. TEST CASES

Test Case	Input	Expected Result	Status
TC 01	20	Valid age: 20	✓ Pass
TC 02	-5	Invalid age message	✓ Pass

Test Case	Input	Expected Result	Status
TC 03	150	Invalid age message	✓ Pass

TEST CASE 01 — NORMAL INPUT

Input

20

Output

Enter age: 20
Valid age: 20

Result: ✓ The program accepts the valid age successfully.

TEST CASE 02 — NEGATIVE AGE

Input

-5

Output

Enter age: -5
Invalid age: Age must be between 0 and 120.

Result: ✓ InvalidAgeException is thrown and handled correctly.

TEST CASE 03 — UNREALISTICALLY HIGH AGE

Input

150

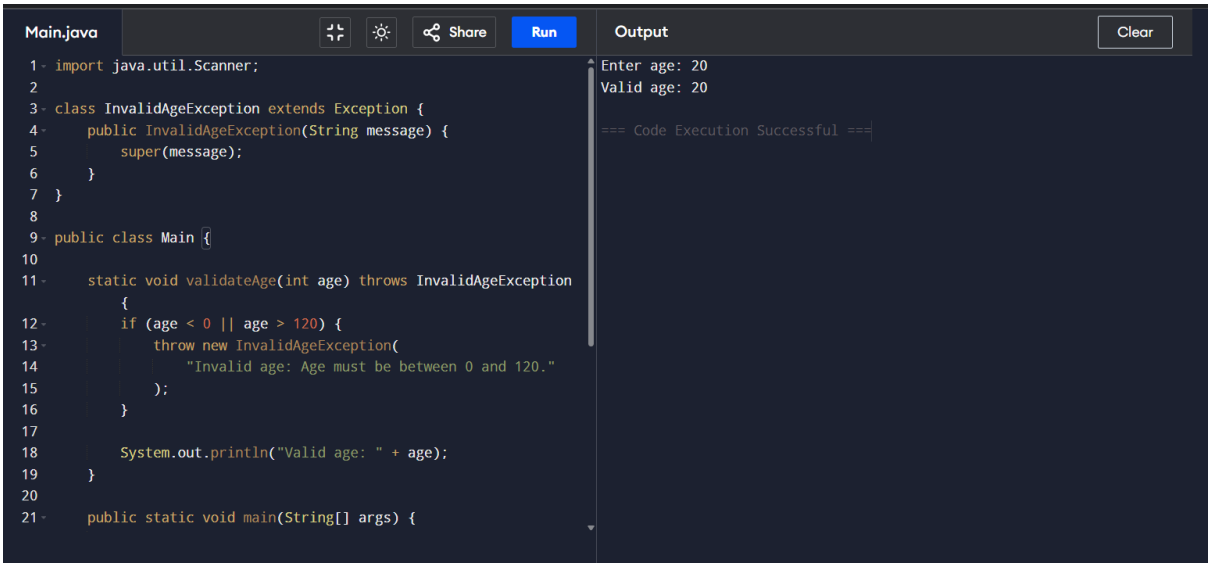
Output

Enter age: 150
Invalid age: Age must be between 0 and 120.

Result: ✓ InvalidAgeException is thrown and handled correctly.

5. EXECUTION SCREENSHOTS

◈ Screenshot – Test Case 01



Input: 20

◈ Screenshot – Test Case 02

Input: -5

Main.java

```java
import java.util.Scanner;

class InvalidAgeException extends Exception {
    public InvalidAgeException(String message) {
        super(message);
    }
}

public class Main {

    static void validateAge(int age) throws InvalidAgeException
        {
        if (age < 0 || age > 120) {
            throw new InvalidAgeException(
                "Invalid age: Age must be between 0 and 120."
            );
        }

        System.out.println("Valid age: " + age);
    }

    public static void main(String[] args) {
```

Output

```
Enter age: -5
Invalid age: Age must be between 0 and 120.

=== Code Execution Successful ===
```

◈ Screenshot – Test Case 03

Input: 150

Main.java

```java
                "Invalid age: Age must be between 0 and 120."
            );
        }

        System.out.println("Valid age: " + age);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter age: ");
        int age = sc.nextInt();

        try {
            validateAge(age);
        } catch (InvalidAgeException e) {
            System.out.println(e.getMessage());
        }

        sc.close();
    }
}
```

Output

```
Enter age: 150
Invalid age: Age must be between 0 and 120.

=== Code Execution Successful ===
```

6. RESULT

The Java program was successfully implemented using a user-defined exception named InvalidAgeException.

The program correctly validates the entered age and throws the custom exception when the age is negative or greater than 120. The exception is successfully caught using the try-catch mechanism and the appropriate message is displayed.

★ RESULT: PROGRAM EXECUTED SUCCESSFULLY ★

7. REFLECTION

Through this assignment, I learned how to create and implement a custom exception in Java. I understood the purpose of throw, throws, and try-catch in exception handling.

The main challenge was understanding how a user-defined exception is created and communicated between methods. By implementing the InvalidAgeException class separately, I was able to handle invalid input in a structured and meaningful way.

This program helped me understand how exception handling can make Java programs more reliable, readable, and user-friendly.

8. DECLARATION

I hereby declare that this assignment is my own work. Any AI or external tool assistance used during the preparation of this assignment has been appropriately disclosed.

Student Name: Geona Alis J

Register No.: 192511088

Signature: Geona

Date: 10/08/2026